

Transformers

Theory, Intuition, and Building from Scratch

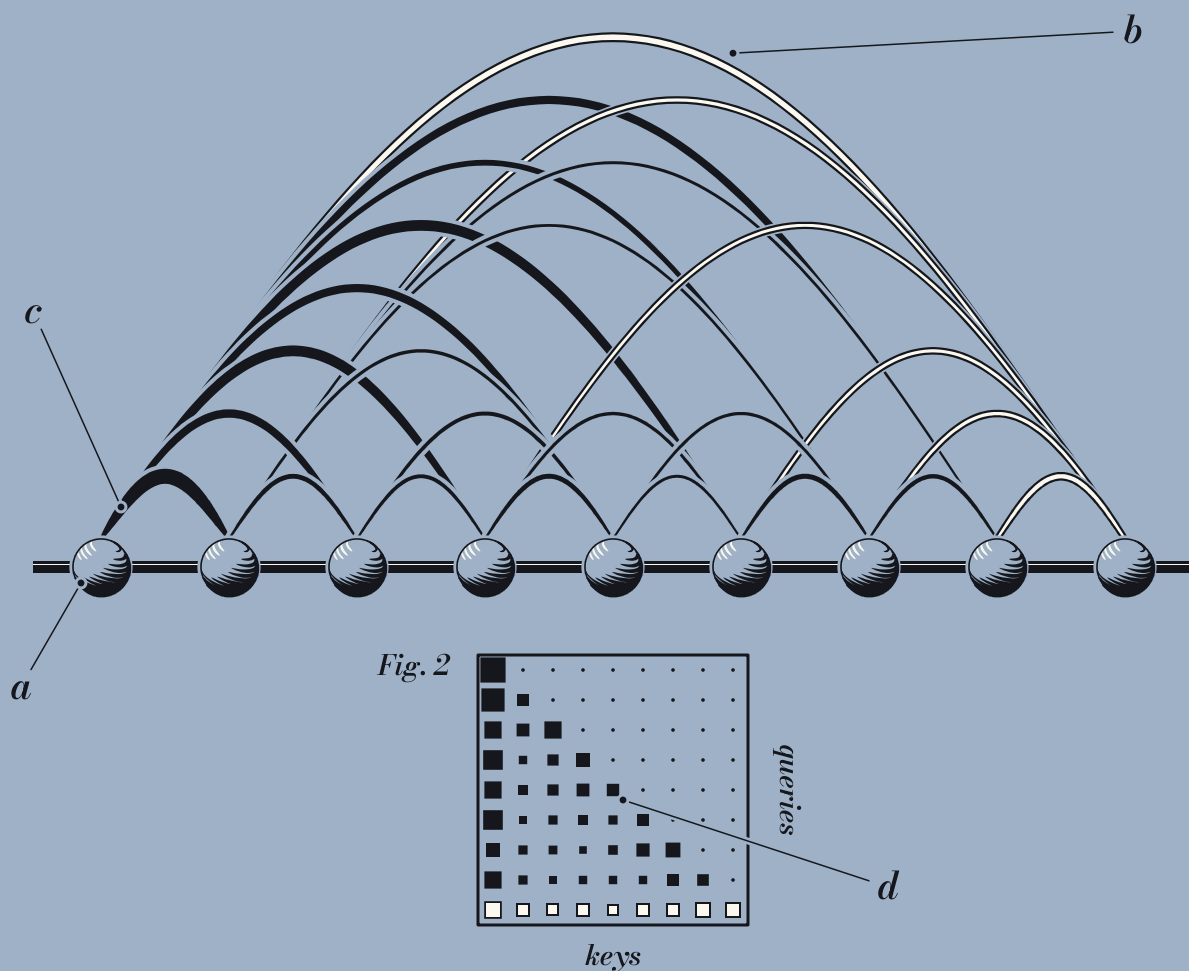


Fig. 1. One causal attention head over nine tokens, each arch weighted by $\text{softmax}(QK^T/\sqrt{d})$. **a** tokens; **b** the last query, in white; **c** the attention sink. **Fig. 2.** The weights as a matrix; **d** its diagonal.